

CHRISTINA DURÓN

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🌐 <https://cduron.info>

ACADEMIC EMPLOYMENT

Assistant Professor of Mathematics

Natural Science Division of Seaver College, Pepperdine University

Aug 2022 – Present

Postdoctoral Research Associate

Department of Mathematics, University of Arizona

Aug 2019 – May 2022

High School Teacher

Mathematics Department, The Webb Schools of California

Aug 2013 – June 2019

RESEARCH INTERESTS

Network Theory; Network Dynamics; Statistical Analysis and Modeling of Complex Networks; Mathematical-Biology

EDUCATION

Claremont Graduate University

Ph.D. in Mathematics

May 2019

- **Thesis:** The Distribution of Betweenness Centrality in Exponential Random Graph Models
- **Advisors:** Dr. Ami Radunskaya (Professor, Pomona College) and Dr. Johanna Hardin (Professor, Pomona College)

University of Washington

Master in Science in Applied Mathematics

June 2013

Swarthmore College

Bachelor of Arts in Mathematics; Minor in Computer Science

May 2012

PUBLICATIONS *

12. Durón C, Kravitz H, Brio M. (2025). Edge percolation centrality: A new measure to quantify the influence of edges during percolation in networks. *PLoS One* 20(9): e0331475. doi: [10.1371/journal.pone.0331475](https://doi.org/10.1371/journal.pone.0331475)
11. Durón C, Kravitz H, Brio M. (2025). Kolmogorov–Smirnov-Based Edge Centrality Measure for Metric Graphs. *Dynamics*, 5(2), 16. doi: [10.3390/dynamics5020016](https://doi.org/10.3390/dynamics5020016)
10. Pfeffer D, Durón C. (2025). An Unlikely Duo: Injecting Art Projects in the Mathematics Classroom. *Journal of Humanistic Mathematics*, 15(1), 87-108. doi: [10.5642/jhummath.RDDV3141](https://doi.org/10.5642/jhummath.RDDV3141)
9. Kravitz H, Durón C, Brio M. (2024). A Coupled Spatial-Network Model: A Mathematical Framework for Applications in Epidemiology. *Bulletin of Mathematical Biology*, 86(132): 1 – 35. doi: [10.1007/s11538-024-01364-3](https://doi.org/10.1007/s11538-024-01364-3)
8. Durón C, Swansen B[†], Kravitz H. (2024). Difference Approximation. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08440](https://doi.org/10.1002/9781118445112.stat08440)
7. Sullivant N, Durón C, Pfeffer D. (2023). From Mirrors to Wallpapers: A Virtual Math Circle Module on Symmetry. *Journal of Math Circles*: Vol. 3: Iss. 1, Article 1. Available at <https://digitalcommons.cwu.edu/mathcirclesjournal/vol3/iss1/1>.

* Authors are ordered by contribution.

[†] Undergraduate student.

6. **Durón C.** (2022). Adaptive Quadrature. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08388](https://doi.org/10.1002/9781118445112.stat08388)
5. **Durón C.**, Farrell A. (2022). A Mean-Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model. *Bulletin of Mathematical Biology*, 84(7): 1 – 19. doi: [10.1007/s11538-022-01026-2](https://doi.org/10.1007/s11538-022-01026-2)
4. **Durón C.** (2021). Linear Algebra, Computational. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat00459.pub2](https://doi.org/10.1002/9781118445112.stat00459.pub2)
3. **Durón C.** (2020). Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks. *PLoS ONE*, 15(7): e0235690. doi: [10.1371/journal.pone.0235690](https://doi.org/10.1371/journal.pone.0235690)
2. **Durón C.**, Pan Y, Gutmann D.H., Hardin J, & Radunskaya A. (2018). Variability of Betweenness Centrality and Its Effect on Identifying Essential Genes. *Bulletin of Mathematical Biology*, 81(9): 3655 – 3673. doi: [10.1007/s11538-018-0526-z](https://doi.org/10.1007/s11538-018-0526-z)
1. Pan Y, **Durón C.**, Bush E.C., et al. (2018). Graph Complexity Analysis Identifies an ETV5 Tumor-Specific Network in Human and Murine Low-Grade Glioma. *PLoS ONE*, 13(5): e0190001. doi: [10.1371/journal.pone.0190001](https://doi.org/10.1371/journal.pone.0190001)

ACCEPTED *

1. Kravitz H, **Durón C.**, Nieves B, Brio M. Data-drive optimization and parameter estimation for a metric graph with application to an epidemic model of COVID-19 in Poland. *An International Journal of Optimization and Control: Theories & Applications*.

UNDER REVISION *

2. O'Brien E, **Durón C.** The Wasserstein metric as a tool to assess burn-in and convergence. *Statistical Science*.
1. **Durón C.**, Mello J[†], Kamps K[†]. Network Evaluation of Influential Sensors on Critical Infrastructures. *The Journal of Dam Safety*.

RESEARCH POSITIONS

Graduate Research Assistant

Pomona College

Jan 2017 – June 2018

- NIH (1R01-CA195692-01) funding under Dr. Ami Radunskaya and Dr. Johanna Hardin

Jet Propulsion Laboratory (JPL) Intern

California Institute of Technology

June 2015 – August 2015

- Implemented the Extended Kalman Filter (EFK) and incorporated inter-robot measurements to improve the state estimation and localization of autonomous vehicles

Mathematical and Theoretical Biology Institute Researcher

Arizona State University

June 2011 – July 2011

- Developed a mathematical model for the evaluation and analysis of the air pollution in Los Angeles

* Authors are ordered by contribution.

† Undergraduate student.

TEACHING EXPERIENCE

Instructor of Record

Pepperdine University

- **MATH 450: Mathematical Statistics** *Spring 2024*
- **MATH 350: Mathematical Probability** *Fall 2025, Fall 2023*
- **MATH 260: Linear Algebra** *Fall 2024, Spring 2023*
- **MATH 150: Calculus I** *Fall 2025, Fall 2024,
Spring 2024, Fall 2023
Spring 2023, Fall 2022*

Instructor of Record

University of Arizona

- **MATH 491: Undergraduate Teaching Assistantship (UTA) Seminar** *Fall 2021 – Spring 2022*
- **MATH 196M: Calculus I Supplementary Seminar** *Spring 2022*
- **MATH 396L: Wildcats Proofs Workshop** *Spring 2022*
- **MATH 464: Theory of Probability** *Fall 2021*
- **MATH 363: Introduction to Statistical Methods** *Spring 2021*
- **MATH 129: Calculus II** *Fall 2020*
- **MATH 475A: Mathematical Principles of Numerical Analysis** *Fall 2020*
- **MATH 163: Basic Statistics** *Spring 2020*
- **MATH 122B: First Semester Calculus** *Fall 2019*
- **MATH 196L: Precalculus Supplementary Seminar** *Fall 2019*

Instructor of Record

The Webb Schools of California

- **Advanced Placement Computer Science Principles** *Fall 2018 – Spring 2019*
- **Honors Precalculus** *Fall 2014 – Spring 2019*
- **Introduction to Computer Programming with Python** *Fall 2014 – Spring 2018*
- **Precalculus** *Fall 2013 – Spring 2019*
- **Integrated Mathematics 2** *Fall 2013 – Spring 2014*

PRESENTATIONS

Student-Directed

3. **Network Evaluation of Influential Sensors: A Proposed Approach** *March 2024*
Pepperdine University; Seaver College Research and Scholarly Research Symposium
2. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** *Jan 2024*
JMM: AMS - PME Undergraduate Student Poster Session
1. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** *July 2023*
Pepperdine University, Natural Science Division Summer Research

Contributed

7. **A Coupled Spatial-Network Model for Epidemiology** July 2024
Frankfurt School of Finance and Management Workshop on Metric Networks
6. **Math Circles: Elevating Student Engagement and Mathematical Enjoyment Outside the Classroom** Jan 2024
University of Arizona Mathematics Educator Appreciation Day Conference (MEAD)
5. **Incorporating Undergraduate Students in Research on Assessing Dam Safety Using Network Analysis** Jan 2024
JMM Special Session: Navigating the Benefits and Challenges of Mentoring Students in Data-Driven Undergraduate Research Projects
4. **Tiling a Chessboard: A Problem Adapted for the Virtual Math Circle** Jan 2023
JMM Special Session: Math Circle Activities as a Gateway into Mathematics
3. **A Mean Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model** May 2021
Los Alamos-Arizona Days Conference (Virtual Poster)
2. **Identifying Super-Spreader Nodes in Scale-Free Networks using Network Centrality Measures** Sept 2020
Arizona Postdoctoral Research Conference (Virtual Talk)
1. **Identifying Treatment Targets for Pediatric Gliomas using Network Centrality Measures** June 2020
SIAM Conference on the Life Sciences (Virtual Talk)

Seminar

4. **A Coupled Spatial-Network Model for Epidemiology** Nov 2023
University of Florida, Health Systems Medicine Seminar (Virtual Talk)
3. **Network Centrality: Theory to Applications** Oct 2021
Arizona State University, Mathematical Biology Seminar (Virtual Talk)
2. **Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks** Feb 2021
Claremont Colleges and University of Utah, Joint Applied Mathematics Seminar (Virtual Talk)
1. **Network Data Analysis Techniques on DESeq and RNASeq Data** Nov 2019
University of Arizona, TRIPODS Research Working Group 6 - Analyzing large-scale point-set data

Industry

1. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** March 2024
California Department of Water Resources
2. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** Nov 2023
Southern California Edison, Dam & Public Safety Group

PROFESSIONAL DEVELOPMENT

Program

- **Project NExT** (Green 2023) Aug 2023 – Aug 2024
Mathematical Association of America (MAA)
 - Focused on all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities

[‡] Presented with undergraduate students.

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Certification

- **Diversity, Equity, and Inclusion in the Workplace** May 2021
University of South Florida
 - Focused on ways that organizations can create a more diverse workplace, address equity issues, and foster inclusivity
- **Effective Online Discussions** June 2020
University of Arizona
 - Developed strategies for designing and facilitating effective online discussions that deepen learning, expand student exposure to curriculum, and increase student engagement
- **Teaching the Large Online Course** June 2020
University of Arizona
 - Developed instructional practices for encouraging student engagement and motivation in a large online class, as well as for effectively managing administrative tasks such as monitoring student progress and conducting assessments

Workshops

- **Workshop on Metric Networks** July 2024
Frankfurt School of Finance and Management
- **Broadening Participation: 2024 Mathematical and Physical Sciences Workshop for New Investigators** July 2024
National Science Foundation
- **Network Modeling for Epidemics** Aug 2020
University of Washington
- **BioBridge Clinic** Jan 2020
University of California, Irvine
- **Computational Genomics Summer Institute** May 2019
University of California, Los Angeles

SERVICE

Mentoring and Advising

- **Undergraduate Research Advisor** (Team of 3 undergraduates) Fall 2025 – Present
Pepperdine University
- **Tidy Thursday Advisor** Fall 2025
Pepperdine University
- **Undergraduate Research Advisor** (Team of 2 undergraduates) Summer 2023 – Fall 2024
Pepperdine University
- **Math Club Vice-Principal Faculty Advisor** Spring 2023 – Fall 2023
Pepperdine University
- **Undergraduate Student Mentor** Fall 2025, Spring 2023 – Fall 2023
Pepperdine University, Faculty/Student Mentor Program
- **Undergraduate Student Mentor** Fall 2021 – Spring 2022
University of Arizona, Women in Science and Engineering (WISE) Program
- **Graduate Student Mentor** Fall 2021 – Spring 2022
University of Arizona, Association for Women in Mathematics (AWM) Mentor Network
- **Undergraduate Research Advisor** Fall 2020 – Spring 2021
University of Arizona

- **Mathematics Undergraduate Teaching Assistantship (UTA) Program Mentor** *Fall 2020 – Spring 2021*
University of Arizona
- **Mathematical Modeling Group Mentor** (Team of 4 undergraduates) *Spring 2020*
University of Arizona
- **Math Club Advisor** *Fall 2017 – Spring 2019*
The Webb Schools of California

University and Departmental Service

- **Creator and principal faculty advisor for Tidy Thursday** *Fall 2024 – Present*
Pepperdine University
 - An unofficial student club focused on exploring and analyzing datasets in **R** to improve data analysis and visualization skills.
- **Institutional Review Board, Natural Science Representative** *Fall 2023 – Present*
Pepperdine University
 - Review research study protocol/application materials (exempt, expedited, and full review) and evaluate them from the perspective of the regulatory criteria for approval.
- **Admissions and Scholarship Committee** *Fall 2025 – Present, Fall 2023 – Fall 2024*
Pepperdine University
 - Review student applications and select Faculty-Staff Scholars based on merit, including academic performance and recommendations.
- **Co-Instructor of RISE/Connection Chapel** *Fall 2025, Fall 2023 – Fall 2024*
Pepperdine University
 - Facilitated discussions on different aspects of science from the Christian perspective, primarily in the disciplines of history, physics, astronomy, and mathematics, with students over five sessions
- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 24/25)** *Fall 2023 – Spring 2024*
Pepperdine University
 - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations
- **Member of Hiring Committee, Visiting Assistant Professor of Mathematics (AY 23/24)** *Summer 2023*
Pepperdine University
 - Served on the five-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom, collecting and consolidating information about the interviews, and making final recommendations
- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 23/24)** *Fall 2022 – Spring 2023*
Pepperdine University
 - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations
- **Member of Review Committee, Excellence in Postdoctoral Mentoring Award** *Spring 2022*
University of Arizona
 - Served on the three-member review committee that determined the recipient of the 2022 Excellence in Postdoctoral Mentoring Award

- **President, Postdoctoral Governance** *Fall 2021 – Spring 2022*
University of Arizona
 - Serve as an in-between for the postdocs and the Postdoctoral Committee, and organize the postdoctoral professional development seminar topics and panels
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Director** *Fall 2021 – Spring 2022*
University of Arizona
 - Coordinate the mentorship of the UTA's, and run the weekly professional development seminar
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Co-Director** *Fall 2020 – Spring 2021*
University of Arizona
 - Supported the Director of the UTA Program, and was responsible for additional duties related to the weekly professional development seminar
- **Vice President, Postdoctoral Governance** *Fall 2020 – Spring 2021*
University of Arizona
 - Supported the President of the Postdoctoral Governance, and was responsible for additional duties related to the postdoctoral professional development seminars
- **Non-Academic Liaison, Postdoctoral Governance** *Spring 2020*
University of Arizona
 - Organized a panel pertaining to non-academic careers for the postdoctoral professional development seminar

Service to the Discipline

- **JMM 2024, Co-Organizer of MAA Project NExT Session** *Jan 2024*
Pepperdine University
 - “Making Student Thinking Visible with Team-Based Inquiry Learning”
- **MAA MathFest 2022, Co-Organizer of Themed Contributed Paper Session** *Aug 2022*
University of Arizona
 - “Math Circles: Talks about Mathematical Joy, Inspirations, and Data-Driven Lessons Learned”
- **Reviewer for:**
 - Complexity *March 2023*
 - Indian Journal of Discrete Mathematics *Nov 2020*
 - DNA and Cell Biology *Jan 2020*
 - Revista de Matemática: Teoría y Aplicaciones *Oct 2019*
- **Mathematics and MATLAB Summer Workshop, Co-Coordinator** *June 2016 – June 2018*
Claremont Graduate University
- **Mathematics and MATLAB Summer Workshop, Co-Instructor** *June 2016 – June 2017*
Claremont Graduate University

Outreach

- **How I Found My Network: My Path to Mathematics** *Nov 2021*
Arizona State University
 - Keynote address for Sonia Kovalevsky Day

- **Tucson Math Circle** Aug 2019 – Present
University of Arizona
 - Co-develop materials and co-run the university sponsored weekly program designed to get middle school students excited about mathematics through hands-on exploration and discovery
- **Association for Women in Mathematics (AWM): Sonia Kovalevsky Day** April 2021
University of Arizona
 - Developed materials and co-ran a workshop designed to bolster female high school and middle school students' passion and enthusiasm for mathematics in a supportive environment
- **The Seven Bridges of Königsberg** Nov 2022
Pepperdine University
 - Talk given to Math Club during Tuesday Tea
- **Using Network Centrality Measures to Identify Unknown Regulatory Pathways in Pediatric Glioma** Sept 2020
University of Arizona
 - Talk given to The MathCats Club (undergraduate math club)

HONORS AND AWARDS

- The Teaching and Service Award** April 2022
University of Arizona, Department of Mathematics
- The Jean E. Miller Excellence in Teaching Award** June 2018
The Webb Schools of California
- The Thompson and Vivian Webb Excellence in Teaching Award** June 2015
The Webb Schools of California
- The Heinrich W. Brinkmann Mathematics Prize** June 2012
Swarthmore College

FUNDING

Research Grants

- **Academic Year Undergraduate Research Initiative** (\$1,000) Sept 2025, Sept 2023
Pepperdine University
- **Collaborative Research Grant for Postdocs** (\$1,500) June 2020
University of Arizona

Travel Awards

- **AIM SQuaRE Research Collaboration** April 2025
American Institute of Mathematics (AIM)
- **TDA-BIO** (\$1,000) Oct 2016
ACM Conference on Bioinformatics, Computational Biology, and Health Informatics

Fellowships

- **Clinic Fellowship** (\$900) Jan 2020
University of California, Irvine
- **Daniel Pick Fellowship** (\$10,000) Oct 2017
Claremont Graduate University

- **Joseph and Elizabeth Peeler Endowed Fellowship** (\$32,570)
Claremont Graduate University
- **CGU Mathematics Fellowship** (\$13,700)
Claremont Graduate University
- **CGU Minority Fellowship** (\$2,000)
Claremont Graduate University

Aug 2015 – June 2017

Aug 2014 – June 2015, June 2017

Aug 2014 – June 2016

SKILLS

Programming Languages

- C (*Moderate proficiency*)
- C++ (*Moderate proficiency*)

- MATLAB (*Proficient*)
- Python (*Proficient*)
- R (*Proficient*)

Scientific Applications

- GitHub

- LaTeX
- RSweave